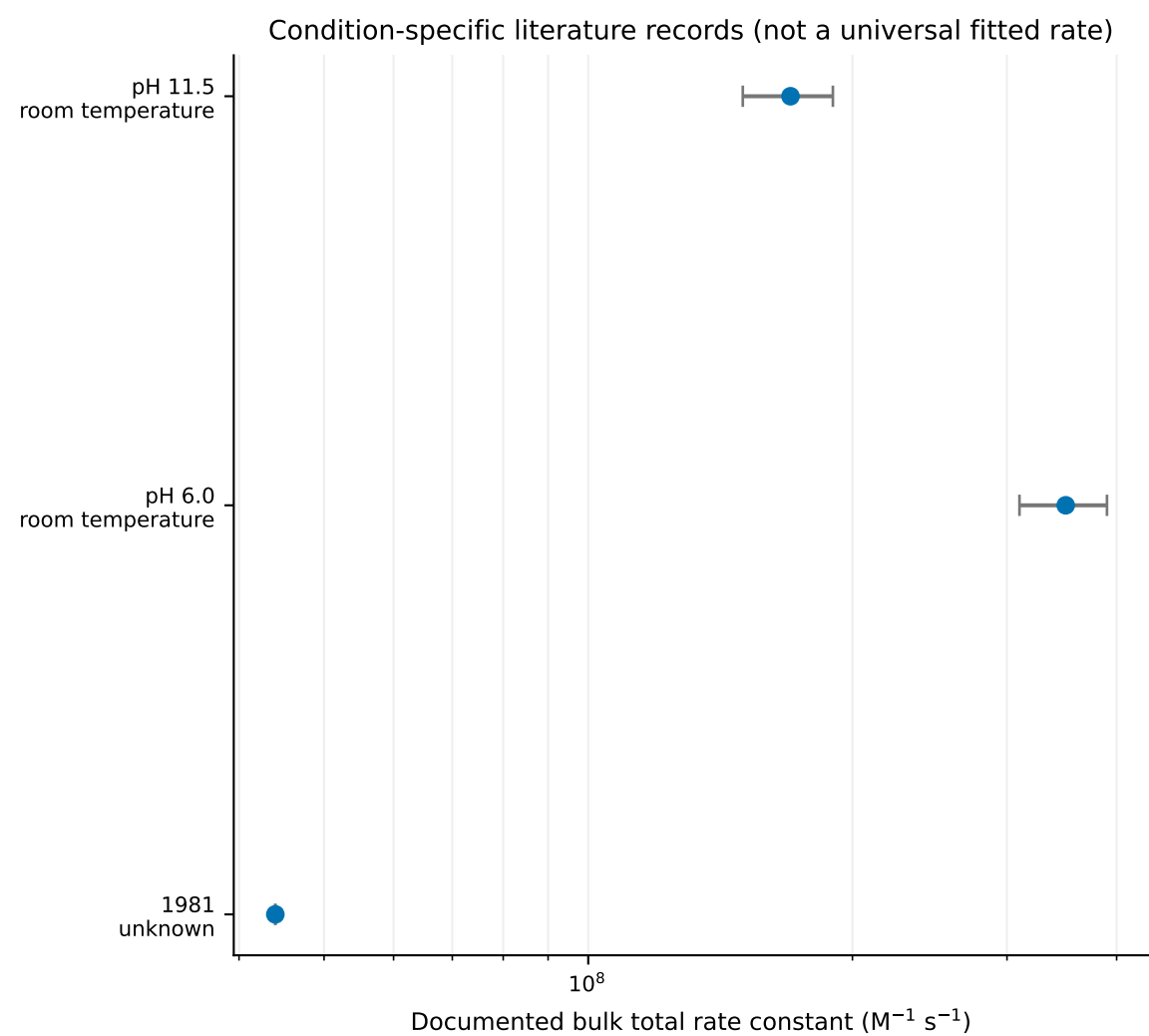




Records remain conditionally distinct

| Record | Conditions/environment                                                                                              | Species/protonation                                                                                                                                                         | Source/allowed use                                                                                                                                                            |
|--------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1981   | pH unknown; unknown; aqueous pulse radiolysis                                                                       | adriamycin semiquinone plus ground-state triplet O <sub>2</sub> ; protonation: unknown                                                                                      | B. A. Svingen and G. Powis, Archives of Biochemistry and Biophysics 209(1), 119-126 (1981), doi:10.1016/0003-9861(81)90263-0; allowed: literature_arithmetic;sensitivity_only |
| pH6    | pH 6.0; room temperature; aqueous phosphate/formate; O <sub>2</sub> or O <sub>2</sub> /N <sub>2</sub> purged        | adriamycin semiquinone plus neutral ground-state triplet O <sub>2</sub> ; protonation: biological-pH internally protonated/zwitterionic semiquinone; O <sub>2</sub> neutral | Land et al. (1985), doi:10.1038/bjc.1985.74; allowed: literature_arithmetic;condition_matched_prediction;sensitivity_only                                                     |
| pH11_5 | pH 11.5; room temperature; aqueous borate/OH <sup>-</sup> ; O <sub>2</sub> or O <sub>2</sub> /N <sub>2</sub> purged | deprotonated adriamycin semiquinone plus neutral ground-state triplet O <sub>2</sub> ; protonation: deprotonated semiquinone; O <sub>2</sub> neutral                        | Land et al. (1985), doi:10.1038/bjc.1985.74; allowed: literature_arithmetic;condition_matched_prediction;sensitivity_only                                                     |